

Counterparts

CIAN DORR

Draft of 12th March 2015
Comments welcome

Contents

1	A-theories and B-theories	1
1.1	Two key questions about time and modality	1
1.2	Propositions	13
1.3	Worlds and instants	24
1.4	Elitism and egalitarianism	32
1.5	Non-factualism and Relativism	38
1.6	Other A-theoretic themes	48
1.7	Summary and preview	53
2	Arguments about A- and B-theories	55
2.8	The argument from spacetime	55
3	Counterpart theory and the fixity of the qualitative	87
3.1	Basic counterpart theory	87
3.2	Some puzzles for Lewis	95
3.3	Contingency-inducing quantification as <i>de re</i> restricted quantification	105
3.4	An exegetical interlude	114
3.5	Alternative accounts of contingency-inducing quantification	118
3.6	Necessary and eternal existence	129
4	Counterpairings, worlds and instants	136
4.1	Multiply <i>de re</i> counterpart theory: two bad ideas	136
4.2	Three better ideas	144
4.3	Constraints on counterpairings	151
4.4	The necessity of identity	161
4.5	Worlds and instants in counterpart theory	170
4.6	Actuality	184
4.7	Counterpart theories are A-theories	187

5	Modal counterpart theory	200
5.1	The generality of modal counterpart theory	200
5.2	Island universes and constraints on counterparthood	209
5.3	Surprising dependencies	223
5.4	What is special about metaphysical modality?	243
5.5	Beyond counterpart theory	248
6	Temporal counterpart theory	253
6.1	Spacetime and the ontology of material objects	253
6.2	Fission and linearity	267
6.3	The modal profiles of instants	271
6.4	Time travel	271
6.5	Relativity <TK>	278

Chapter 2

Arguments about A- and B-theories

2.8 The argument from spacetime

The weightiest argument for the B-theory of time is based on a certain picture of the fundamental structure of the world inspired by modern spacetime physics. This picture can be captured in a slogan: *The world is a four-dimensional spacetime manifold*.

To get the argument into focus, we will need to try to capture the content of the manifold picture in more precise terms. In doing this we will have to make various choices, both about the general metaphysical question how, in general, one should understand the enterprise of ‘characterising the fundamental structure of the world’, and about how much specific content to build into this particular account of the fundamental structure of the world. Many of these choices will make little or no difference as far as the argument for the temporal B-theory is concerned. But for now, let’s not worry about making the argument needlessly specific: if we can get a workable argument onto the table, we can consider afterwards how its premises might be weakened.

Here is a first stab at spelling out the manifold picture:

The Manifold Picture: Concrete reality consists, fundamentally speaking, of *spacetime points* and *spacetime regions*. The structure of concrete reality is fully captured by a four-dimensional *differentiable structure* (a criterion for what counts as a “smooth” co-ordinate system) over the set of spacetime points, together with a short list of ‘physically distinguished’ scalar, vector and/or tensor fields (e.g. the metric field, the electromagnetic field, the mass-density field) on this differentiable structure.

This formulation helps itself to some heavy-duty metaphysical jargon which does not wear its meaning on its sleeve—‘fundamentally speaking’ and

the notion of ‘fully capturing structure’. The task of finding a disciplined regimentation of this kind of talk is an important one, which has been a focus for much recent discussion. But for present purposes, what we need to do is to extract from the manifold picture some consequences expressed in terms of concepts we have already been working with, notably the notion of metaphysical necessity and that of naturalness.

Here is one important claim which can be plausibly taken to follow from the manifold picture:

Manifold Supervenience: All true propositions follow with metaphysical necessity from (i) truths about which spacetime points and regions there are; (ii) truths to the effect that a certain point or region is part of a certain other region; (iii) truths about which co-ordinate systems over the set of spacetime points are smooth; (iv) the truth, for each physically distinguished tensor field and spacetime point, about which tensor at that point is the value of that physically distinguished field; and (v) a “that’s all” truth.

The “that’s all” truth referred to in (v) is supposed to be a proposition which somehow captures the idea that there isn’t any more to the world than those spacetime points and/or regions, and that there isn’t any more to the structure of the world than is captured by truths of kinds (ii) and (iii). Of course, the task of explaining what *that* means raises the very same conceptual difficulties as the task of making the manifold picture precise. Nevertheless, modal formulations like Manifold Supervenience are useful to work with, since we often don’t need to know exactly how to properly articulate a “that’s all” truth in order to make judgments about what it can accomplish.

(Those who think they understand a non-trivial ‘factually’ operator may deny that Manifold Supervenience follows from the manifold picture. It would be natural for them to insist that even if the world is fundamentally just a spacetime manifold, only the *factual* true propositions need to be entailed by the facts about spacetime and its fields, while non-factual true propositions can vary independently. But it’s worth noting that for the purposes of establishing certain conclusions with a B-theoretic flavour, the result of weakening Manifold Supervenience by replacing ‘All true propositions’ with ‘All factual true propositions’ would be just as good as Manifold Supervenience itself.)

Another kind of claim that seems to be implicit in the manifold picture

involves the notion of naturalness:

Manifold Naturalness: The only natural classes of points and regions are those picked out by simple conditions on the physically distinguished fields.

Manifold Naturalness allows us to say, for example, that the set of points where the electromagnetic field vector is zero is a natural class, and that the set of points where the scalar curvature field is positive is a natural class. It is supposed to rule out the idea that there are some 'natural joints' in space-time that are completely uninteresting from the standpoint of the physically distinguished fields. This is plausibly part of the kind of 'completeness' that the manifold picture claims on behalf of those fields.

Manifold Naturalness talks about the naturalness of classes rather than that of properties. One might contemplate deriving Manifold Naturalness from the claim that the physically distinguished fields are the only instantiated (perfectly?) natural properties. But this claim brings up special problems. If we want to think of the metric tensor (for example) as a property, it would seem to be a property of *tensors at points*. And such tensors are standardly treated as rather elaborate set-theoretic constructions. For example, the metric tensor at a point would normally be identified by a certain function from pairs of functions from points to real numbers to real numbers. The idea that non-mathematical properties of such mathematical entities are perfectly natural seems to embody a rather heavyweight form of Platonism. It also raises worries about arbitrariness, since there are many different constructions which can equally well do the job of a tensor at a point (cf. Benacerraf 1965). For this reason, there is good reason to retreat from the claim that the physically distinguished fields are themselves perfectly natural properties to the less committal claim that the perfectly natural properties and relations, whatever they might be, are such that the facts about their pattern of instantiation can be read off the facts about the differentiable structure and the physically distinguished fields in some reasonably simple way. This leaves us free to entertain many different ideas about what these perfectly natural properties and relations actually are, and how the facts about the physically distinguished tensor fields supervene on them.

Have we done enough by way of articulating the manifold picture to make it clear why it might seem to require a B-theory of time? Not yet. For we haven't said anything about what the 'physically distinguished' fields *are*, or what kind of pattern they make. For all we have said, it might be

that, for example, one of the physically distinguished fields is a 'degree of futurity' scalar field, such that the points where the value of this field is zero constitute a natural three-dimensional slice through spacetime which could play the role of the present time (the infamous 'moving spotlight'). There is an intelligible version of the A-theory of time which accepts Manifold Supervenience and Manifold Naturalness while positing such a field, and claiming that it is metaphysically necessary that the propositions that *will* be true, or *have been* true, are all and only those that *would* be true in a world where the values of the degree-of-futurity field at each point were decreased, or increased, by a constant amount.¹ But the picture we are trying to get on the table is not supposed to contain anything like that! To make this clear, let us understand the view to include the following claim:

Manifold Homogeneity: None of the especially natural classes of spacetime points is a connected, spacelike, three-dimensional spacetime region.

(The notion of a region's being 'spacelike' is one that can be understood in terms of spacetime's geometric structure. In both relativistic and Neo-Newtonian spacetime, the relevant kind of structure can be captured by a physically distinguished metric tensor.)

Even this isn't enough to rule out some other kinds of spacetime-friendly A-theory. Some *growing block theorists* may be willing to accept that reality consists, fundamentally, of a spacetime manifold endowed with certain physical fields. It's just that they think that standard B-theorists are wrong about the *extent* of this manifold: rather than going on forever in the 'futurewards' direction, or ending in a 'Big Crunch' at which some of the physical fields increase without limit, the manifold just *cuts off* some fourteen billion years after the Big Bang. There is a consistent version of this view which accepts Manifold Supervenience: according to it, it is metaphysically necessary that a proposition was, or will be true just in case it would be true in a possible world similar to the actual world in certain specified respects but where the extent of spacetime is smaller or larger.² Similarly, a 'shrinking block' theorist might think of reality as a spacetime manifold truncated in

¹There is also a closely related but arguably more economical view which makes do with a fundamental property of 'presentness' attaching to spacetime points, defining the degree of futurity of a point as the spatiotemporal distance between it and the hypersurface of present points, multiplied by minus one if the point is *before* this hypersurface. For an able defence of this kind of view, see **DeasyMST**

²According to the most common version of the growing block view, spacetime is not a manifold in the strict sense but rather a *manifold with boundary*: the result of stitching

the other direction. And even weirder views are conceivable, such as a view that accepts spacetime points corresponding to the past and the future while denying the existence of a 3d slice corresponding to the present. But none of these shapes is the shape of spacetime according to the picture we are trying to get on the table. According to that picture, the spacetime manifold is “as big as it can be” consistent with the physical fields being mathematically well-behaved. Let us make this explicit by setting down Manifold Completeness as part of our official statement of the manifold picture:

Manifold Completeness: There is no way to embed the spacetime manifold with its physically distinguished fields into a larger connected manifold of the same dimensionality on which all the physically distinguished fields are everywhere smooth.

Now that we have filled in some of the details in the manifold picture, let us see what we can do to get the argument from the picture to the temporal B-theory into focus. I can see two broad lines of argument, the first of which targets propositional eternalism, and the second of which targets temporal egalitarianism. The first argument can be spelled out as follows:

The Supervenience Argument:

- S1 The manifold picture is true.
- S2 If the manifold picture is true, then if any propositions are temporarily true, some propositions about the values of the physically distinguished fields at particular spacetime points are temporarily true.
- S3 If the manifold picture is true, then all true propositions about the values of the physically distinguished fields at particular spacetime points are eternally true.

Conclusion: There are no temporarily true propositions.

together a three-dimensional manifold (the present) with a four-dimensional one (the past). This view is inconsistent with Manifold Homogeneity, since the class of boundary points is certainly a natural one. However, I see no good reason why the growing block theorist should not, instead, think of the ‘forward edge’ of reality as topologically *open* rather than closed. On such a view, all spacetime points are ‘past’ insofar as it makes sense to apply temporal classifications to spacetime points at all. But that need no more prevent proponents of the view from agreeing that all kinds of things are the case *now* than it need prevent them from agreeing that all kinds of things will be the case *later*.

The second argument is as follows:

The Homogeneity Argument:

- H1 The manifold picture is true.
- H2 If the manifold picture is true, then instants of time are connected, spacelike, three-dimensional regions of spacetime.
- H3 If the manifold picture is true, then no connected, spacelike, three dimensional region of spacetime has an especially natural unit set.

Conclusion: The unit set of the present time is not especially natural.

Let us continue to postpone the question what reason there might be for believing the manifold picture, and consider what can be said on behalf of the remaining premises of these arguments.

S2 can be motivated by appeal to the claim that metaphysical necessity implies eternity (Dorr and Goodman n.d.). Thus, if a proposition follows from some eternally true propositions with metaphysical necessity, that proposition must itself be eternally true. But according to Manifold Supervenience (which we have built into the manifold picture) all true propositions follow with metaphysical necessity from truths about which spacetime points there are, about which co-ordinate systems are smooth, about the values of the physically distinguished fields, and a 'that's all' truth. So, if there are temporarily true propositions, there must be temporarily true propositions in at least one of these categories. But it is hard to see how there could be temporarily true propositions about *which spacetime points or regions there are*, or about the smoothness of co-ordinate systems, without there also being temporarily true propositions about the values of the physical fields at points—what counts as a tensor depends on the underlying differentiable structure. And even without an account of what the 'that's all' truth amounts to, it is hard to see how *it* could be the one source of temporariness. If all the spacetime points and regions that there *are* are things that exist eternally, it is utterly implausible that there *used to be*, or that there *will be*. And it is even more implausible that there *used to be*, or *will be*, more 'kinds of structure' in spacetime than there now are (let alone that the truth about how this extra structure worked or will work is somehow metaphysically necessitated by the truth about the structure that there is now). Thus, if there are any temporary truths in the supervenience base provided by Manifold

Supervenience, at least some of it are surely truths of class (iv), about the values of the physically distinguished fields.

(In the form in which I have stated it, S2 will be rejected by non-factualists who restrict Manifold Supervenience to factual truths. For them, the interesting argument in the vicinity of the Supervenience Argument is one in which the quantification in Manifold Supervenience and in the conclusion is restricted to factual propositions.)

The motivation for S3 is just that it seems quite hard to think of any plausible answer to the question *which* truths about, e.g., the electromagnetic field are temporary, especially given the constraint that the truths about the future and past configurations of that field must be necessitated by truths of the kinds listed in Manifold Supervenience. Of course, if our list of physically distinguished included some kind of ‘degree of futurity’ scalar field, there would be a plausible story about how *its* values on particular spacetime points have been different or will be different, namely that when the degree-of-futurity of a spacetime point p is x , it *will* be the case for every $y < x$, and *has* been the case for every $y > x$, that the degree-of-futurity of p is y . But Manifold Homogeneity rules this out.³ If we thought that the manifold was in some way “truncated”, we might be able to figure out a story about how its present structure necessitates truths about its future growth or shrinkage, which would make for change even in the overall configuration of familiar physical fields like the electromagnetic field. But given Manifold Completeness, the plausible lines of investigation in this direction are blocked. Note that I am *not* claiming that there is anything *incoherent* about the idea that total configuration of the physical fields over the four-dimensional manifold should change—one can in principle imagine A-theoretic views on which reality consists of a four-dimensional manifold undergoing all kinds of changes. The problem is that given what the manifold picture says about the *actual* configuration of the fields, it is hard to think of a plausible account of how that configuration could necessitate any non-trivial truths about how the configuration has been or will be different.

Let us turn next to the Homogeneity Argument. We guaranteed premise H3 by building Manifold Homogeneity into the manifold picture, so the action is in premise H2. This premise has been almost universally taken

³Certain physically non-standard accounts of the configuration of the standard physical fields—for instance, a view that posits a special slice through spacetime on which these fields fail to be continuous and take value zero—might open up room for similar stories, but these are also hard to reconcile with Manifold Homogeneity.

for granted in discussions of time and spacetime. (*cites!*) This practice is encouraged by standard introductions to the concept of spacetime in physics and philosophy, which emphasise the analogy between the role played by spacetime points (in physical theories that talk about spacetime) and the role played in Newtonian physics by ordered pairs comprising a point of absolute space and an instant of time. But we don't want the label 'spacetime' to carry any argumentative weight: if calling the manifold 'spacetime' forced a certain view about how its parts relate to instants of time, we could just call it something else (for example, we could use Forrest's rather catchy term 'the all-pervading aether' **cite**) without changing our picture of its structure in any way. Nevertheless, there is a strong *prima facie* case for H2, namely that given the basic elements provided by the manifold picture, it is hard to think of any *other* good candidates to identify with the instants of time. Even if we help ourselves to the resources of set theory or some even richer domain of 'constructed' objects, no *interestingly* different candidates to be identified with instants of time jump out at us. (There are many *uninterestingly* different candidates, since there are many possible set-theoretic constructions which correspond in some natural way with regions of the relevant kind; but these don't help with the argument, since Manifold Homogeneity entails that none of *them* are especially natural either.) It won't do, for example, to simply announce that instants of time are real numbers, and that the present instant of time is the real number 0, and leave it at that. For we need our metaphysics of instants to fit together with our metaphysics of ordinary objects in such a way as to yield answers to questions like 'What is it for an object to be spherical at an instant?'. And whatever objects might be, it is obscure what it could mean for one of them to be spherical 'at' a real number.

Thus, there is a strong *prima facie* case that A-theorists must reject the manifold picture — it is little wonder that the literature tends to run something like the manifold picture (or some mild generalisation thereof) together with the B-theory under the label 'four-dimensionalism'. Some A-theorists will take a radical stand against the manifold picture, claiming that spacetime physics, while perhaps useful as a predictive tool, is radically misleading as a guide to fundamental ontology. Other A-theorists will attempt to keep as much of the manifold picture as they can, making only the minimum alterations required to block the arguments above. Most obviously, they might reject Manifold Homogeneity while keeping all the rest of the picture,

expanding the list of 'physically distinguished' fields to include something like a "degree of presentness" field, or some other structure which picks out a natural slice through spacetime. Call this the 'Supplemented Manifold Picture'.

As accounts of the fundamental structure of reality go, the Supplemented Manifold Picture seems rather suspicious. If there are good grounds for positing the extra structure that privileges a special three-dimensional slice, they must be extremely different in character from the grounds that support positing, for example, the structure induced by the electromagnetic field. Considered on its own terms, apart from its putative role in explaining the truth of certain descriptions of the world in non-fundamental terms, the privileging of a special three-dimensional slice detracts markedly from the simplicity and elegance of the whole package. The ordinary physical fields fit together in lovely ways. There are simple laws which characterise their interrelations; and the laws for any one field take account of each of the others. Dropping any one of the physical fields would make for a less simple and unified package, since it would leave us without an explanation for certain patterns in the remaining fields. By contrast, the Supplemented Manifold's special slice-privileging structure is a 'nomological dangler'. Mentioning it does not help you to describe the patterns in the other fields. The structure is not something you would be tempted to posit, unless you were convinced by a philosophical theory that told you that some such posit was required in order to provide an adequate supervenience base for the familiar empirical facts.

The proponent of the Supplemented Manifold will respond, of course, that a world that lacked this special structure could not be *our* world. Facts about the location of the privileged slice play a crucial role in the supervenience basis for any non-eternal truth, such as the truth that I am sitting. But we should be on our guard when philosophers come along with controversial accounts of what it takes for familiar truths to obtain, according to which accommodating these truths requires us to make our overall theory of the fundamental structure of the world more complicated than it would otherwise be. Very often, the right response to philosophers in this dialectical position is that their account of the familiar facts is mistaken, and that on a proper account of these facts, their obtaining requires no special-purpose additions to the fundamental structure of the world. An uncontroversial example: some benighted philosopher might claim that we can *just see* that certain objects are *at rest*, and maintain that in order to account for this obvious

fact, we must enrich our picture of the fundamental structure of spacetime to include a standard of absolute motion and rest. We respond that a correct metaphysical account of the obvious facts, informed by a correct semantic theory of ‘at rest’ that recognises its vagueness and context-sensitivity, will show that they can obtain even in a world where spacetime lacks this kind of structure. A more controversial example: some dualists claim that the facts about consciousness show that physics is incomplete as an account of the fundamental structure of the world; in some sequences of cases where physics sees only continuous variation, there are sharp natural boundaries between the cases where the “lights of consciousness” are on and the cases where they are off. Physicalists like me reply that a correct account of the obvious facts about consciousness, informed by a correct semantic theory of ‘conscious’ that recognises its vagueness, shows that no such additional natural structure is required for these facts to obtain.⁴ Of course, considerations of what is required to provide an adequate supervenience basis for the obvious empirical facts are not *irrelevant* to the investigation of the fundamental structure of the world—physics is an experimental discipline! But a lot of the time, we seem to get on better by looking for accounts of the fundamental structure of the world that look simple and unified taken on their own terms, while keeping a very open mind about what might be required to provide an adequate supervenience base for familiar facts. By these standards, the Supplemented Manifold’s natural slice through spacetime looks like a unwelcome philosophical imposition.

For the sake of concreteness, I have chosen to focus on a rather specific account of the fundamental structure of the world, which is controversial for many reasons that have nothing to do with the philosophy of time. But many of the details of the manifold picture, and of our ways of making it precise are quite irrelevant to the philosophical dialectic. For the Supervenience Argument, all that mattered about the picture was the fact that there is a certain body of truths in which it is hard to see how there could

⁴For good measure, here is an *extremely* controversial example that fits the same pattern. Many philosophers of quantum mechanics (*cite Maudlin*) claim that obvious facts, such as the fact that a certain pointer ended up pointing to a certain number, show that the unitarily evolving wavefunction is incomplete or incorrect as an account of the fundamental structure of the world, and must either be supplemented (with ‘hidden variables’) or revised (with ‘collapses’). Many-worlds theorists like (*cite Wallace*) reply that a correct account of the obvious facts in question shows that they require no such supplementation or revision.

be temporariness, and which, according to it, necessitate every truth. For the Homogeneity Argument, all that mattered was that the only obvious candidates the picture provides play the role of *instants of time* are entities none of which has an especially natural unit class. There are many ways in which we could modify the picture without affecting these features:

- We could introduce further natural properties and relations on the spacetime points and regions that go beyond the kind of structure that is naturally captured in the language of tensor fields. For example, one could use a binary “belongs to the backwards light cone of” relation among spacetime points to capture the manifold’s geometric structure. This introduces a fundamental asymmetry that is not present when the only relevant structure is the metric tensor field.⁵
- We could reduce the fundamental ontology, perhaps by eliminating regions and keeping only the points, or by eliminating the points and keeping only some special regions (such as the regions with nonzero volume, or the one-dimensional submanifolds).
- We could, more radically, find some other fundamental ontology in terms of which one can capture the mathematical structure of a differential manifold—perhaps even one that does not involve points or regions at all.⁶ Also, we could consider various *generalisations* of the concept of a differentiable manifold that have been suggested as possibly of physical interest (**cites**).
- We could expand the fundamental ontology by adding a category of *particles*, each of which stands in a fundamental ‘occupation’ relation to a certain large collection of points (and/or to the corresponding region), in such a way that the resulting pattern still does not provide any natural way of singling out a three-dimensional subregion. This dualistic ontology of spacetime and particles is one that some metaphysicians may find more familiar than one on which the parts of spacetime are the only fundamental entities. But contrary to the impression one might get from many popularisations, modern ‘particle physics’ does very little to support the claim that particles belong on

⁵**cite Maudlin

⁶See Geroch n.d., Dorr 2011b, Arntzenius and Dorr 2011.

our list of fundamental entities.⁷

- More generally, we could expand the fundamental ontology by adding a category of ‘occupiers’, with a fundamental ‘occupation’ relation setting up a one-to-one correspondence between occupiers and spacetime regions, or between occupiers and those spacetime regions at which the non-geometric fields are nonzero. To make sure that the occupiers aren’t mere dangles in the arrangement, one might also want to say that some of the fundamental properties and relations that determine the facts about the field-values at a given point are properties of the occupiers, or relations some of whose relata are occupiers.⁸ It is hard for me to see what could motivate adding occupiers to the fundamental ontology: they merely make the ontology more complex without bringing any obvious theoretical benefits. Nevertheless, provided that the occupiers are sprinkled around spacetime rather than confined to a single special slice, their addition does not interfere with the Homogeneity Argument. And insofar as hard to see how the facts about the occupation relation that establish the one-one mapping from occupiers onto regions could be temporary, the addition of occupiers also leaves the Supervenience Argument standing.

Even allowing for modifications of these sorts, it is doubtful whether the manifold picture is a true account of the actual fundamental structure of the world. The empirical success of quantum theory seems to require some kind of far-reaching revisions to the kinds of hypotheses about fundamental structure I have been considering, and there is almost nothing I could say that would be uncontroversial about what the resulting picture should look like. My own sense is that it is quite unlikely that the evidence that requires us to replace classical field theory with quantum field theory, or the development of quantum gravity, will have much bearing on the A-theory/B-theory debate. The quantum-theoretic successors of the manifold picture will be *prima facie* just as inhospitable to the A-theory. They too will admit modifications like the Supplemented Manifold which would remove the inhospitability by adding extra fundamental facts which are naturally thought of as changeable and as marking out a new natural division within the classes of entities that look like natural candidates to be the instants. But such modifications will

⁷***Cite Wallace (and others?) on the conditions for the emergence of particle-like behaviour in QFT.***

⁸For a clear example of such dualism, see Heller n.d.

not look any more natural, or more in keeping with the spirit of the theory, than they do in the case of the manifold picture. If I am right about this, we might as well use the manifold picture in its original form as a proxy for whatever more adequate accounts of the structure of the world might emerge from reflection on up-to-date physics.

So far, I have managed to say nothing about relativity theory. As I have spelled it out, the manifold picture is consistent with there being a uniquely natural ‘foliation’ of spacetime (i.e. a division of it into smooth, non-overlapping three-dimensional slices). Pre-relativistic physics can be interpreted as describing a manifold of this sort—what is called ‘Neo-Newtonian spacetime’.⁹ (‘Neo-’ because, by contrast with Newton’s own account of the metaphysical underpinnings of his physics, it provides no *natural* way to make sense of questions like ‘Is this particle in motion or at rest?’ and ‘How fast is it moving?’.) Even if physics turned out to favour Neo-Newtonian spacetime, the imposition of an extra structure picking out a special *slice* from the privileged foliation as “the present” would look suspicious in the way I have been complaining about. However, *relativistic* manifolds have some further features which make the addition of such structure especially alien and unsimple.

In a generic relativistic manifold, there is no simple way to read off a uniquely natural foliation from the geometric structure. The geometric structure of these manifolds *does*, indeed, determine a natural binary relation over spacetime points which we might think of as having the significance of “beforeness”. But by contrast with the Neo-Newtonian case, the natural “beforeness” relation does not induce a natural foliation on the manifold. In Neo-Newtonian spacetime, the relation *x is not before y and y is not before x* is an equivalence relation; its equivalence classes are the leaves of the privileged foliation. In the relativistic case, by contrast, this relation (known as ‘spacelike separation’) fails to be an equivalence relation, since it is not transitive. A point *z* can be spacelike separated from each of two points *x* and *y*, even though *x* is before *y*. (This is illustrated in figure ***.)

Relativistic spacetimes have the following further feature: once we are *given* a particular spacelike hypersurface *S* in the manifold, there is a natural and obvious way of making sense of questions like ‘Is this body in motion or at rest relative to *S*?’ and ‘How fast is this body moving relative to *S*?’ (assuming that “bodies” are things that somehow single out timelike

⁹***cite Sklar, Earman***

lines through the manifold, their “worldlines”). For in relativistic geometry, unlike the geometry of Neo-Newtonian spacetime, it makes good sense to ask about the *angle* at which a (smooth) timelike line intersects a (smooth) spacelike hypersurface. In particular, it makes sense to ask whether a timelike line is “perpendicular” to a spacelike hypersurface at the point where they intersect.¹⁰ When a body’s worldline intersects S perpendicularly, it is natural to say that it is “at rest” relative to S; when it intersects S at an angle, the magnitude of this angle will correspond to the particle’s *speed* relative to S. Thus, if we were to supplement the standard kinds of spacetime structure by adding an extra fundamental property, the ‘glow of presentness’, which attaches to a single spacelike hypersurface in the manifold, we would find ourselves willy-nilly rehabilitating those awkward Newtonian questions about “absolute” motion and rest. This seems a step backwards. The general consensus is that it is a great advantage of Neo-Newtonian spacetime over Newton’s metaphysics of absolute space that it allows us to avoid having to posit such explanatorily irrelevant, and epistemologically intractable, matters of fact. Relativistic spacetime without a ‘glow of presentness’ preserves this advantage, as does Neo-Newtonian spacetime even with a glow of presentness superadded. But relativistic spacetime with a glow of presentness does not.

This contrast must not be overstated. In certain *particular* relativistic manifolds—indeed, in certain large classes of such manifolds—there are non-arbitrary mathematical recipes for extracting a foliation of the manifold from the facts about the physically distinguished fields. To take a simple example, consider a geometrically flat (Minkowskian) manifold that carries some mass-bearing field, whose total mass is finite when integrated over any spacelike hypersurface. In this setting, we can pick out a unique foliation by the condition that relative to each of its leaves, the centre of mass of the universe should be at rest. And in some families of *general*-relativistic spacetimes, we can even find non-arbitrary mathematical recipes for extracting foliations just from the behaviour of the *metric* field, without having to attend to any of the other fields. For example, in ‘Big Bang’ spacetimes with an initial spacelike singularity, we can assign a number to each point that is the length of the longest path running backwards from it to the singularity; if everything goes well, the sets of points that are all assigned the same

¹⁰Be warned, however, that relativistic “angles” do not work at all like ordinary Euclidean angles: on the most natural way of measuring them, the angles at which timelike lines intersect spacelike hypersurfaces can be arbitrary real numbers. ***Cite Maudlin***

number by this function will constitute a foliation of the manifold by smooth spacelike hypersurfaces.¹¹

These facts mean that we have to be careful in explaining the sense in which these relativistic manifolds provide no “privileged” or “absolute” criteria of motion and rest. In the favourable cases, it is possible to introduce a meaning for ‘at rest’ without doing anything as *manifestly* arbitrary as saying ‘I hereby stipulate that things that are at rest relative to the walls of my lab count as being “at rest”’, or ‘I hereby stipulate that the trajectory that passes through spacetime points x and y counts as being “at rest”’. But this is also true in Neo-Newtonian spacetime. If the total mass of the universe is finite, there are good questions about whether any given trajectory is in motion or at rest relative to the centre of mass of the universe, and if we wanted to stipulate that ‘at rest’ should mean ‘at rest relative to the centre of mass of the universe’, there would be no obvious sense in which we had made an *arbitrary* choice. The widespread consensus that Neo-Newtonian spacetime is preferable to Newton’s metaphysics as a setting for pre-relativistic physics need not be based on ignorance of this fact. There really is an epistemologically important difference between the hypothesis that spacetime is Neo-Newtonian and Newton’s metaphysics of absolute space; the point about the centre of mass of the universe just shows we have to be cautious in using words like ‘arbitrary’, ‘conventional’, ‘privileged’, and ‘natural’ in capturing the relevant difference. And the fact remains that the relevant attractive feature of Neo-Newtonian spacetime is one that we give up if we want to supplement a relativistic manifold with the ‘glow of presentness’.

Relativity makes the faults of the Supplemented Manifold theory—its lack of simplicity, its disunified character—worse than they would otherwise be. But my assessment is that the weightiest reasons for resisting theories like the Supplemented Manifold are independent of relativity. They are simply the general reasons for being suspicious whenever philosophers look at some elegant, austere account of the fundamental structure of the world and tell us it *must* be incomplete, on the grounds that it doesn’t provide what philosophical reflection has revealed to be the necessary conditions for the obtaining of some familiar facts. Attempts to draw conclusions

¹¹However, some of these spacetimes admit several different mathematically natural foliations, so that there is still something offputtingly arbitrary and epistemologically problematic about a metaphysical theory that selects one of them as *the* one whose leaves sometimes enjoy the glow of presentness. ***see Belot on CMC time, etc.***

of that sort from philosophical reflection have a terrible track record. So, modesty is called for. Before rejecting any account of fundamental structure on philosophical grounds of this sort, we had better subject the seemingly convincing philosophical reflections to the highest level of scrutiny, on the lookout for loopholes we might have missed.